

VSA + MACHINE LEARNING FINANCIAL ADVISOR BOT

**A transparent XAUUSD M5 advisory system combining
Volume Spread Analysis, Logistic Regression and MetaTrader 5**

Final Project Report Draft

Project template: 4.2 Financial Advisor Bot

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Educational prototype — not financial advice and not a claim of future profitability

Abstract

This report presents a financial advisor prototype for non-technical gold traders. The system converts a precisely specified Volume Spread Analysis (VSA) strategy into candidate setups, then uses supervised machine learning to estimate whether each candidate will reach its labelled objective. It therefore does not attempt to predict every future price movement. It predicts the probability of success for a strategy-defined event. Two required candidate models—Logistic Regression (LR) and Random Forest (RF)—are trained and evaluated separately on the same chronological partitions. LR is selected using development-set F1 (0.515 versus 0.508), then evaluated once on an unseen test set where it records F1 0.556. A rule-only baseline reaches a higher raw test F1 because it accepts every candidate, but produces a lower simulated trading result: 4.208R compared with 7.374R for VSA+LR. A matched MetaTrader 5 replay gives VSA+LR a 2.85% return and 1.31% maximum equity drawdown, compared with 0.23% and 1.69% for VSA-only. The web interface explains VSA context, model probability and feature contributions; a numerical audit reconstructs all 65 unseen LR probabilities with maximum error 1.67×10^{-16} . An MT5 CSV bridge provides alerts while demo execution remains disabled by default. Results are encouraging but limited by 431 extracted setups, broker tick volume, uncertain transaction costs, and incomplete target-user evaluation.

Keywords

Financial advisor bot; Volume Spread Analysis; Logistic Regression; explainable AI; XAUUSD; MetaTrader 5; Streamlit; chronological evaluation.

Table of Contents

Chapter 1 — Introduction	3
Chapter 2 — Literature Review	6
Chapter 3 — Design	10
Chapter 4 — Implementation	15
Chapter 5 — Evaluation	21
Chapter 6 — Conclusion and Future Work	27
References	29
Appendices	31

Chapter 1 — Introduction

1.1 Project concept and template

Retail trading platforms make complex market data easy to access, but access is not the same as understanding. A beginner can see candlesticks, volume and automated strategies yet still struggle to identify why a setup matters, how uncertainty should be expressed, or how much capital should be exposed. This project addresses that gap through a financial advisor bot for XAUUSD (gold against the US dollar) on five-minute data. The advisor combines a formalised Volume Spread Analysis strategy with machine learning, a web explanation layer and a MetaTrader 5 (MT5) integration. Its purpose is decision support and education, not guaranteed profit or unrestricted autonomous trading.

The work follows University of London project template 4.2, Financial Advisor Bot. The template expects financial-data analysis, an AI/ML workflow, an interface usable by non-technical users, tested advice and an integrated prototype. The implementation meets these expectations by separating offline data preparation and training from the runtime advisor. Python extracts and labels VSA candidates, trains LR and RF independently, selects a model using development data and saves evidence. Streamlit presents recommendations and explanations. MT5 exports completed bars and consumes a narrow CSV signal contract. This division avoids retraining when a user opens the website and makes each component testable.

1.2 Personal and user motivation

My motivation comes from approximately three years of personal interest in trading. At the beginning, my learning was not properly structured. Like many beginners, I was attracted by the possibility of extra income and encountered signal groups, shortcuts and unrealistic promises before receiving a clear explanation of risk management, stop-loss placement or why a setup might be valid. This experience showed me that access to a trading signal is not the same as understanding a trading decision.

The intended users are beginners, including students and people in full-time work, who cannot watch charts for many hours. They may copy social-media signals, paid groups or automated bots without understanding candle behaviour, confirmation, news risk or the amount they could lose. The project was therefore designed as a responsible monitoring and learning tool. It identifies a strategy-defined opportunity, estimates its historical probability and explains the setup,

uncertainty and risk controls. It is not intended to give blind directions or promise profit.

1.3 Problem statement

The central problem is not simply whether the next candle rises or falls. The strategy first defines a meaningful event—a VSA change-in-behaviour (CAB) candle and one of three confirmation scenarios—and the model estimates whether that event is likely to achieve the labelled outcome. This is a conditional prediction problem. It is also imbalanced: only 144 of 431 extracted setups are labelled positive. Accuracy alone could therefore reward a model that mainly rejects trades. The project uses precision, recall, F1 and balanced accuracy alongside trading and risk measures.

The second problem is trust. A probability without an account of its origin can encourage automation bias. The project distinguishes four layers: the VSA domain explanation, the LR model explanation, the action recommendation and the safety controls. Logistic Regression is attractive because each standardised input contributes directly to the log-odds, making local numerical reconstruction possible. This is more faithful than attaching a generic natural-language reason after prediction, although it remains an association rather than a causal explanation.

1.4 Aim, objectives and research questions

The aim is to design, implement and critically evaluate a transparent VSA-based machine-learning financial advisor for beginner XAUUSD traders.

- Formalise the three mirrored VSA scenarios, thresholds, stop-loss rules and 70/30 trade management so that Python and MT5 implement the same specification.
- Create a reproducible XAUUSD M5 setup dataset from two years of completed bars and prevent temporal leakage through chronological, embargoed partitions.
- Train LR and RF separately, store their artefacts and results separately, compare them under equivalent conditions, and justify the selected advisor.
- Provide faithful LR explanations, accessible domain explanations, calibrated safety language and a non-technical Streamlit interface.
- Integrate the advisor with MT5 through a validated CSV bridge and evaluate software behaviour, predictive performance, simulated trading and robustness.

The report asks: RQ1—does a learned filter improve a fixed VSA strategy on unseen chronological data? RQ2—does LR or RF provide the better development-set balance between predictive performance and explainability? RQ3—can the

selected model's outputs be explained faithfully and communicated safely? RQ4—does the integrated system operate reliably across Python, Streamlit and MT5?

1.5 Scope, contribution and ethical position

The scope is deliberately narrow: XAUUSD, M5 bars, tick volume, three VSA scenarios, binary outcome prediction, LR and RF as the required comparison, and advisory/demo integration. The originality lies in joining a trader-defined CAB state machine to a transparent event-quality model, then carrying the same evidence into a beginner-facing explanation and an MT5 bridge. The project does not claim that VSA is universally profitable, that tick volume equals centralised traded volume, or that a historical result predicts future performance. Live-money automation, multi-asset deployment and reinforcement learning remain outside the evaluated scope.

The system communicates educational status, probability, risk and rejection states. Demo trading is disabled by default; stale signals are rejected; one-position and age controls are available; and the news filter can block recommendations. The target-user study requires informed consent. At the date of this draft the information sheet, questionnaire and analysis workbook exist, but signed forms and genuine responses have intentionally not been invented. They will be appended after collection.

Chapter 2 — Literature Review

2.1 Technical analysis, volume and VSA

Weak-form market efficiency proposes that current prices already reflect information in past prices, challenging systematic returns from chart rules (Fama, 1970). Later empirical work complicates a simple dismissal. Brock, Lakonishok and LeBaron (1992) found that moving-average and trading-range rules contained information in the Dow Jones series, while Lo, Mamaysky and Wang (2000) argued that technical patterns can be formalised and tested rather than treated only as visual folklore. These studies do not validate any specific VSA strategy. They do, however, justify converting discretionary rules into explicit algorithms and evaluating them out of sample.

Volume can add information that is absent from price alone. Blume, Easley and O'Hara (1994) developed a model in which market statistics that include volume can improve inference about information quality. The current project uses that idea cautiously. XAUUSD in MT5 commonly supplies broker tick volume rather than consolidated exchange volume. Tick volume records quote activity and is therefore a proxy whose meaning depends on the broker feed. A CAB candle is consequently treated as an operational feature—relative activity against the previous 30 bars—not as direct evidence of institutional accumulation or distribution.

Volume Spread Analysis is popularised in practitioner literature as the joint reading of volume, spread and close location (Coulling, 2013). Practitioner sources are useful for eliciting the domain strategy but provide less reliable evidence of profitability than peer-reviewed tests. The project therefore separates domain provenance from statistical evidence. The user supplied the CAB and three scenarios; the software defines every threshold; and performance claims are based only on the project's held-out data. This addresses the examiner's request to reduce the number of concepts and validate a smaller core rigorously.

2.2 From trading rules to supervised learning

A rule-based bot and a machine-learning advisor answer different questions. The VSA state machine determines whether a candidate structurally exists. The classifier estimates the conditional probability that this candidate reaches its labelled objective. This hybrid design preserves the strategy while allowing historical cases to influence acceptance. It also provides a meaningful baseline: the same VSA candidates can be replayed with and without the learned filter.

Logistic Regression models the log-odds of a binary response as a linear combination of features (Cox, 1958). With standardisation and regularisation, it is computationally efficient and its coefficient structure supports direct explanation. Its limitation is structural: nonlinear interactions must be represented explicitly, and correlated features can make individual coefficients unstable. A model may therefore be transparent yet still poorly specified.

Random Forests average many decision trees grown from resampled data and feature subsets, reducing the variance of an individual tree and capturing nonlinear interactions (Breiman, 2001). They can perform well without a linear decision surface but are harder to explain at the level of a single recommendation. Global impurity importance is not a faithful local explanation and can be biased. The examiner's concern that RF is difficult to explain is therefore valid, but it is not a reason to omit RF. Comparing it under the same split tests whether added complexity produces enough benefit to justify its interpretability cost.

Large-scale finance studies show that flexible ML can reveal nonlinear relationships, but gains depend on careful regularisation and evaluation (Gu, Kelly and Xiu, 2020). This project is much smaller and does not reproduce asset-pricing research. The relevant lesson is methodological: model complexity must be earned by out-of-sample evidence. A HistGradientBoosting classifier is included as contextual evidence, not as the deployment decision. Its development F1 of 0.524 is only marginally above LR's 0.515 on 65 rows, while its unseen-test F1 is 0.542 versus LR's 0.556. The result does not support replacing the transparent candidate.

Table 1. Critical comparison of model families used in the study.

Approach	Strength in this project	Main limitation	Decision
VSA rules only	Exact domain fidelity; clear baseline	Accepts every detected candidate; no learned discrimination	Retained as baseline
Logistic Regression	Direct probability; faithful additive contributions; fast inference	Linear log-odds; sensitive to specification	Selected from required LR/RF candidates
Random Forest	Nonlinear interactions; robust tree ensemble	Local reasoning is less direct; importance is not full XAI	Stored comparison model
Gradient boosting context	Tests whether greater flexibility changes conclusions	Less transparent; small validation advantage may be noise	Context only, not deployed

Source: author's own work unless otherwise stated.

2.3 Evaluation under temporal dependence

Random shuffling is inappropriate when examples occur through time because later information can leak into training and adjacent observations can share market conditions. Time-respecting evaluation is therefore central. Bergmeir, Hyndman and Koo (2018) show that cross-validation assumptions for time-series models require attention to dependence. The present study uses an earlier training block, a later development block for thresholds and model choice, a 180-minute embargo and a final unseen test block. Walk-forward folds provide additional evidence about regime variation.

Backtesting introduces another source of optimism. Trying many indicators, thresholds or periods and reporting the winner can select noise. White's (2000) reality check and the probability-of-backtest-overfitting work of Bailey et al. (2017) demonstrate why repeated search must be treated sceptically. The project limits candidate families, records validation thresholds, preserves the unseen test and reports negative walk-forward folds rather than hiding them. This does not remove selection bias, because the strategy itself was developed before formal evaluation, but it makes the remaining threat explicit.

Class imbalance also changes metric choice. Saito and Rehmsmeier (2015) show that precision–recall views can be more informative than ROC-style summaries when positive cases are rare. The project reports F1, precision, recall and balanced accuracy. F1 is the development selection metric because both missed valid setups and false acceptances matter. Trading outcomes are reported separately: predictive classification and financial utility are related, but neither is a substitute for the other.

2.4 Explainable AI and responsible advice

Explainable AI is not a single property. Barredo Arrieta et al. (2020) distinguish interpretability, post-hoc explanation and audience-dependent requirements. Ribeiro, Singh and Guestrin (2016) likewise emphasise explaining individual predictions rather than trusting model accuracy alone. For this project, a useful explanation must answer three different questions: why the VSA scanner found a scenario; which variables increased or decreased LR probability; and why the system recommends act, watch or avoid.

The deployed explanation is intrinsic to the fitted LR pipeline. For a standardised feature vector z , the log-odds equal the intercept plus the sum of coefficient–feature products. Each local contribution is therefore $z_j\beta_j$. Adding all 24 contributions and the intercept reconstructs the model probability. This is a faithful

computational account. It is not causal: a positive contribution does not prove that changing that input would cause success. It is also not automatically understandable to beginners, so the interface pairs numeric contributions with domain definitions and safety advice.

Random Forest feature importance is retained only as global context. Describing that chart as full XAI would conflate explanation types, exactly as the examiner warned. A later version could add SHAP values, which provide an additive feature-attribution framework for many models (Lundberg and Lee, 2017), followed by a user study to test whether explanations improve comprehension rather than merely appearing technical.

2.5 Web-mediated financial advice and synthesis

A non-technical advisor must reduce cognitive load without presenting uncertainty as certainty. The Streamlit interface therefore separates recommendation, probability, model evidence, VSA narrative, risk plan and limitations. Model training is moved out of the request path; the page loads saved metadata, the current signal and compact result tables. This improves response time and prevents the interface from silently changing the deployed model.

The literature points to a defensible project position. Technical patterns should be specified and tested, not assumed. Volume is informative in theory but broker tick volume is a proxy. Complex models are not automatically superior. Time order and selection bias dominate evaluation design. XAI must be faithful to computation and understandable to its audience. These principles directly shape the architecture, model choice, testing strategy and claims in the following chapters.

Chapter 3 — Design

3.1 Requirements and design principles

The functional requirements are to ingest completed XAUUSD M5 bars, detect mirrored CAB scenarios, derive pre-outcome features, train LR and RF separately, select a model using development evidence, issue an advisory probability, explain that probability, and transfer a validated signal to MT5. Non-functional requirements include reproducibility, low-latency page opening, traceable artefacts, readable explanations, safe defaults and consistent rules across Python and MQL5.

Five design principles follow: separation of training from inference; chronological rather than random evaluation; one canonical strategy specification; evidence saved as CSV/JSON rather than only displayed; and advice bounded by safety gates. The web layer never becomes the source of model truth. It reads artefacts produced by the backend, which makes the recommendation reproducible and the interface faster.

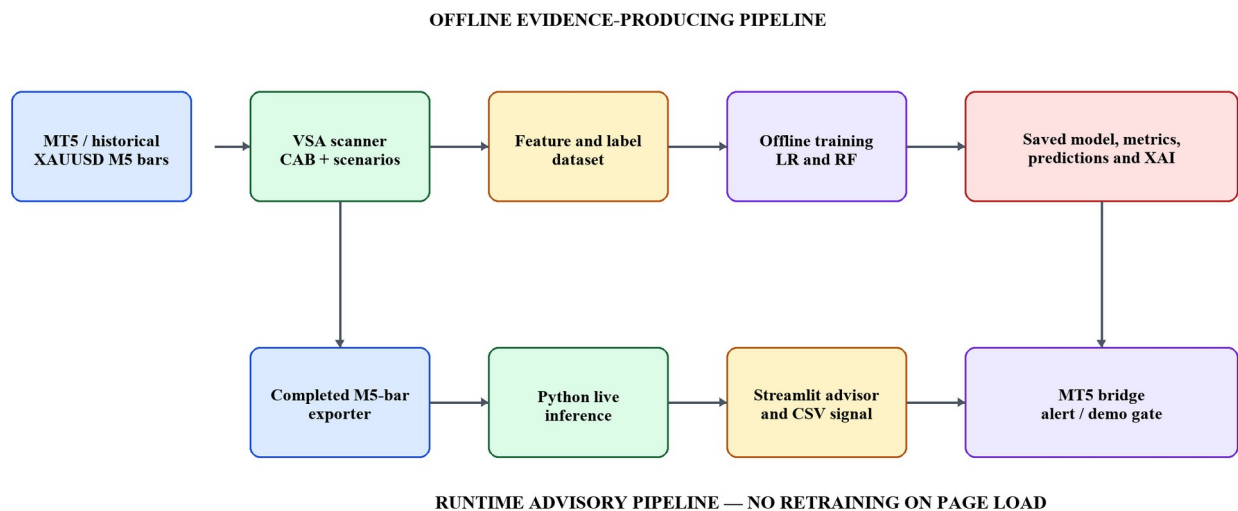


Figure 1. System architecture and data flow. Source: author's own design.

3.2 Formal VSA strategy

A buy-side CAB is a bearish candle occurring after a downward move whose tick volume is at least 1.5 times the maximum/benchmark of the preceding 30 completed bars. Its high is the upper trigger and its low the lower trigger. A valid breakout requires a momentum candle with body-to-range ratio at least 0.60, range at least 1.20 times its recent reference and volume lower than the CAB. Sell logic mirrors every condition around a bullish CAB in an upward context. These fixed

definitions replace ambiguous phrases such as ‘high volume’ and ‘momentum’ and are used by the dataset generator and advisor.

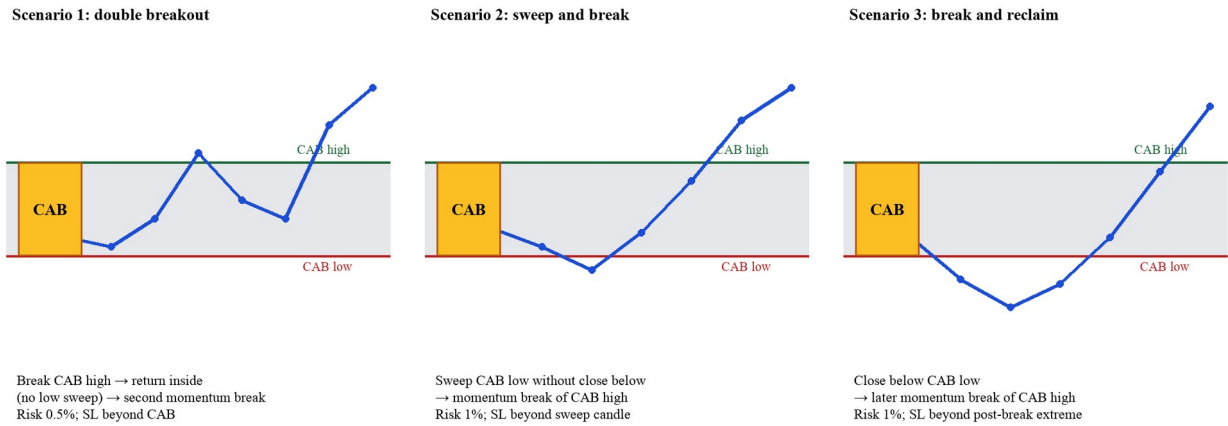


Figure 2. Formalised buy-side VSA scenarios; sell logic is mirrored. Source: author’s own strategy specification.

Table 2. Canonical scenario, risk and exit specification.

Scenario	Confirmation	Risk	Stop-loss anchor	Management
1	First high break, return inside without low sweep, then second lower-volume momentum high break	0.5%	Opposite side of CAB	Close 70% at 1R; move SL to entry; run 30% to 4R or breakeven
2	Sweep—not close—beyond opposite CAB trigger, followed by lower-volume momentum break	1.0%	Beyond the sweep candle	Same 70/30 management
3	Close beyond opposite CAB trigger, then reverse lower-volume momentum break	1.0%	Extreme formed after the CAB break	Same 70/30 management

Source: author’s own work unless otherwise stated.

3.3 Learning problem and features

The unit of learning is a completed VSA setup, not a single market bar. Each row contains only information available at the signal time. The label `valid_trade` is one when the setup reaches the predefined outcome before invalidation under the labelling horizon and zero otherwise. This formulation avoids claiming that the model predicts an unconstrained market future; it estimates strategy-conditional success.

Table 3. Complete deployed 24-feature contract used by LR and RF. Every value is calculated before the outcome label.

#	Feature	Group	Meaning at signal time
1	scenario_id	Domain	Scenario 1, 2 or 3 code

#	Feature	Group	Meaning at signal time
2	direction_num	Domain	Buy = 1; sell = -1
3	relative_spread	CAB	Candle spread against its recent reference
4	relative_volume	Confirmation	Signal-candle volume ratio
5	close_location	Candle	Normalised close position within the range
6	body_ratio	Candle	Body divided by complete candle range
7	upper_wick_ratio	Candle	Upper wick divided by candle range
8	lower_wick_ratio	Candle	Lower wick divided by candle range
9	cab_range	CAB	CAB high minus CAB low
10	cab_relative_volume	CAB	CAB volume relative to earlier bars
11	entry_to_stop_distance	Risk	Planned distance from entry to stop-loss
12	risk_reward_distance	Risk	Target geometry relative to stop distance
13	bars_after_cab	Sequence	Completed bars from CAB to confirmation
14	had_opposite_sweep	Sequence	Whether the opposite trigger was swept
15	breakout_strength	Confirmation	Magnitude of the valid trigger break
16	return_3	Context	Backward-looking three-bar return
17	return_6	Context	Backward-looking six-bar return
18	return_12	Context	Backward-looking twelve-bar return
19	volatility_12	Context	Backward twelve-bar variability
20	volatility_48	Context	Backward forty-eight-bar variability
21	hour	Time	Hour of the completed signal candle
22	day_of_week	Time	Weekday of the signal
23	london_session	Time	London-session indicator flag
24	newyork_session	Time	New York-session indicator flag

Source: author's own work unless otherwise stated.

3.4 Data partition and model-selection protocol

The source dataset contains 143,020 XAUUSD M5 bars from 1 August 2024 to 17 August 2026. The scanner extracts 431 setups: 144 positives and 287 negatives. Setups are ordered by signal time and divided 70/15/15 into 301 training, 65 development and 65 unseen-test rows. A 180-minute embargo separates

development from test to reduce boundary contamination. Scaling and model fitting use training only. Probability thresholds are selected on development only. Test labels are opened once for final comparison.

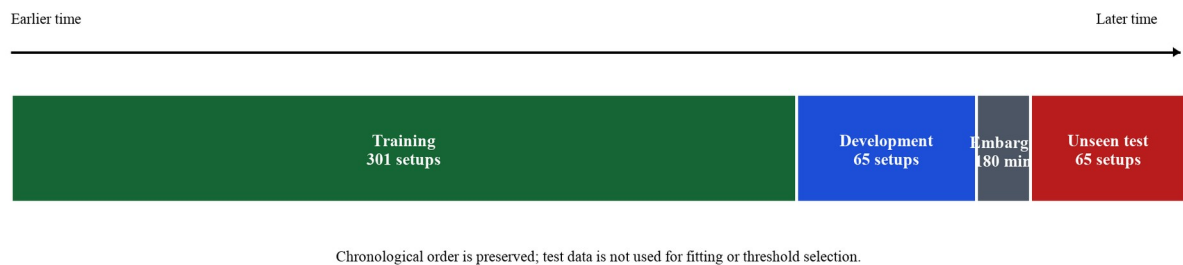


Figure 3. Chronological 70/15/15 partition with embargo. Source: author's own dataset design.

LR and RF are trained as separate pipelines and stored as `lr_vsa_bot.joblib` and `rf_vsa_bot.joblib` with separate metrics, predictions and confusion matrices. The predefined decision rule selects the higher development F1 among these candidates. Explainability is a secondary design criterion: if predictive evidence is approximately equal, LR is preferable because its local decision can be reconstructed exactly. The threshold itself is not a trading certainty; it is a development-tuned operating point.

3.5 Explanation and safety design

The interface presents four explanation layers. The domain layer identifies CAB direction, trigger levels and scenario sequence. The model layer displays the selected probability and positive/negative LR contributions. The action layer maps probability and safety gates to BUY, SELL, WATCH or AVOID. The safety layer shows risk percentage, stop and target geometry, freshness and news state. Keeping these layers separate prevents a domain narrative from being misrepresented as model XAI.

MT5 integration uses a 29-field CSV schema and a file in the MT5 Common Files area. The bridge validates header/data boundaries, symbol, timeframe, signal age and action. Alerts and push notifications are permitted independently of demo execution. `EnableDemoTrading` defaults to false. This fail-closed design allows integration evidence without risking an unintended order.

3.6 Workplan and risk management

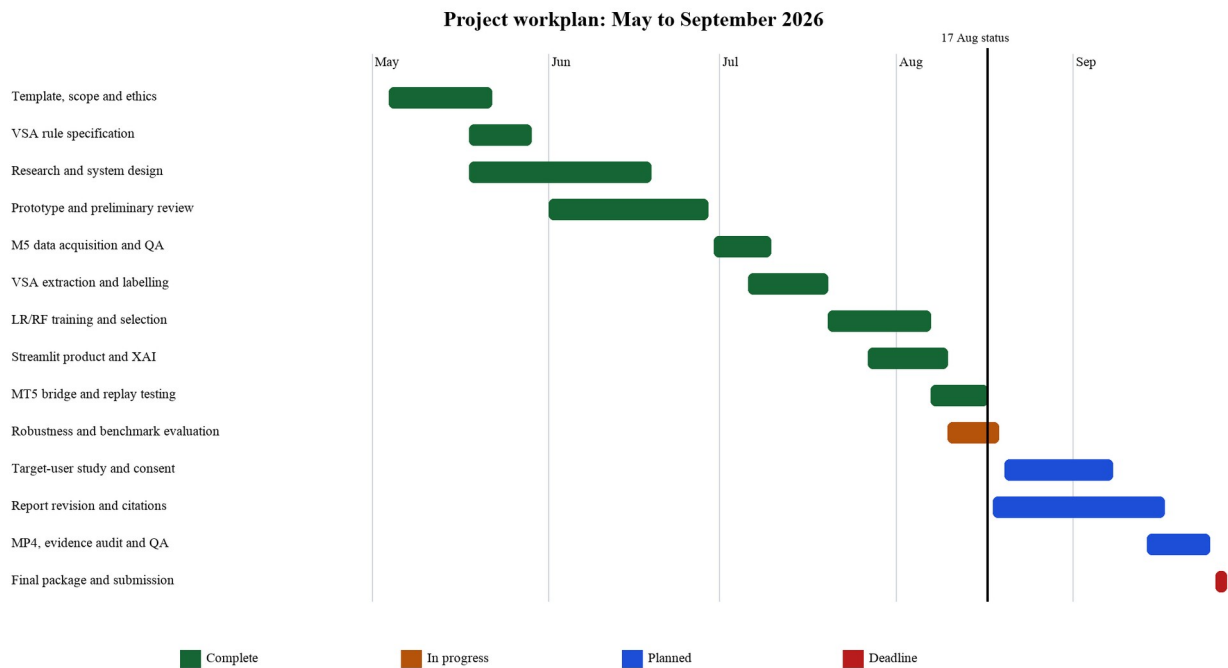


Figure 4. May–September 2026 workplan and Gantt chart. The vertical line is the 17 August status date; the final deadline is 28 September. Source: author’s own project workbook.

The remaining critical path is user evaluation, final report revision, the required MP4 demonstration and submission QA. Key risks are insufficient genuine participants, late evidence capture, unreported data leakage and inconsistency between Python and MT5. Mitigations are a consent-led 10–15 participant study, a fixed evidence index, automated regression tests, matched benchmark settings and a final rule-to-code audit. Profitability is not a completion criterion; defensible evaluation and honest limitations are.

Chapter 4 — Implementation

4.1 Technology stack and repository structure

The prototype is implemented in Python 3 with pandas and NumPy for data processing, scikit-learn for LR/RF pipelines, joblib for artefact persistence, Streamlit for the web interface and pytest for automated tests. MT5 components are written in MQL5. CSV and JSON are used deliberately at the integration boundary because they are inspectable during assessment and supported by MT5 without a network service. The repository separates modules, backend_training, models, results, scripts, mt5, tests and docs.

Training runs through a backend command rather than the Streamlit page. The training script loads completed M5 bars, calls the VSA extractor, creates chronological partitions, fits candidate pipelines, sweeps thresholds on development data, serialises both models, selects LR or RF, evaluates the final test and writes metrics. This directly addresses the earlier performance issue: the ML Advisor page now reads saved model metadata, a current signal and small evidence tables instead of recomputing two years of bars on every page load.

4.2 Strategy extraction and labelling

The scanner is a state machine. A new CAB starts a bounded observation window. Subsequent completed bars update state variables such as first breakout, return inside, sweep, close beyond the opposite trigger and reversal breakout. A setup is emitted only when a scenario reaches its confirmation transition; future bars are used only by the label evaluator, never by signal-time features. Buy and sell branches call mirrored comparisons so that direction does not silently change thresholds.

Algorithm 1 — VSA-based machine-learning advisor pipeline

1. Load and time-sort completed XAUUSD M5 OHLCV bars; reject duplicates and invalid values.
2. For each eligible bar, test trend context, candle direction and CAB relative-volume threshold.
3. Store CAB high/low and advance the mirrored scenario state machine using completed bars only.
4. When Scenario 1, 2 or 3 confirms, calculate signal-time features, entry, scenario stop and 1R/4R targets.

5. For historical construction only, label whether the predefined valid outcome occurs before invalidation.
6. Chronologically split setup rows; fit LR and RF separately on training data.
7. Select each probability threshold on development F1; compare candidates and save the chosen pipeline.
8. At runtime, calculate the same feature vector, predict success probability and create an explanation.
9. Apply freshness, news, threshold and risk gates; export the 29-field CSV and update the web view.
10. MT5 validates the CSV and issues an alert; a demo order is possible only when explicitly enabled.

4.3 Separate LR and RF training

Both candidates receive the same training rows and feature contract. LR uses standardisation inside its pipeline so saved inference applies the training scaler consistently. RF is fitted separately and never borrows LR predictions. Development threshold sweeps produce an LR threshold of 0.35 and RF threshold of 0.46. Model files, validation predictions, test predictions, confusion matrices and metric files are separate, satisfying the teacher's request for independently stored evidence.

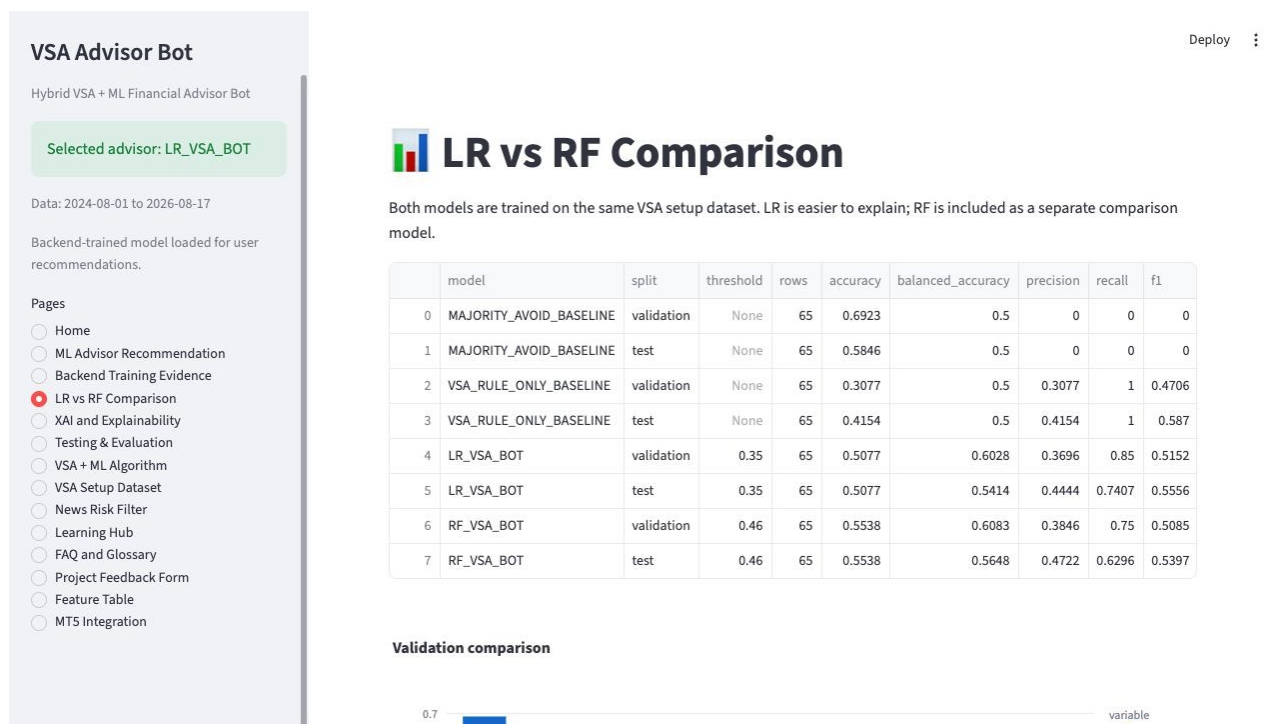


Figure 5. Streamlit LR/RF evidence page showing separately evaluated candidates and baselines. Source: author-captured prototype screen.

4.4 Streamlit website and performance optimisation

The web application contains fourteen pages covering recommendation, backend evidence, LR/RF comparison, XAI, evaluation, algorithm, dataset, news, learning material, glossary, feedback, feature table and MT5 integration. The first view uses plain language and exposes uncertainty rather than showing internal debug structures. Development-status sentences, raw JSON and AI-sounding implementation notes were removed from the user-facing view. Technical evidence remains available on dedicated audit pages.

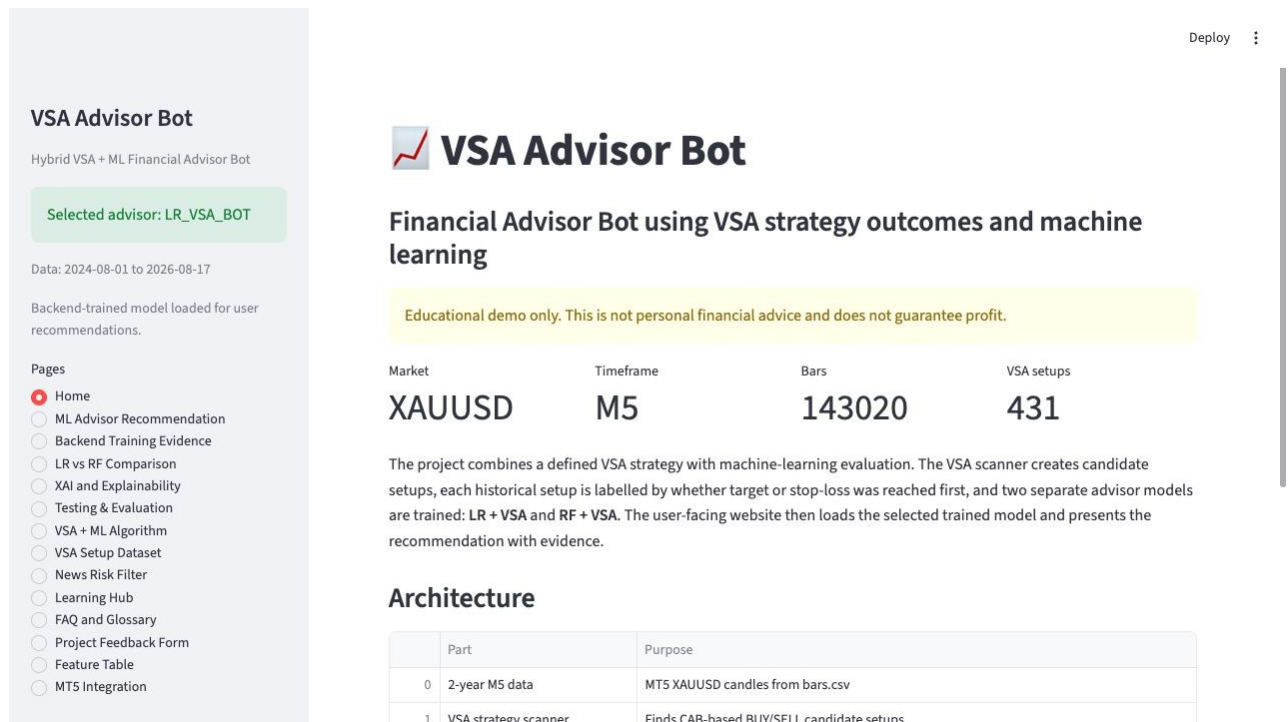


Figure 6. Optimised Streamlit home page and navigation. Source: author-captured prototype screen.

Learning Hub, blog, FAQ and glossary

The website is both an advisor and an educational interface. Its Learning Hub contains twelve short lessons covering candles, tick volume, VSA, the CAB definition, all three scenarios, VSA with ML, LR, RF, news and risk management. A project blog, Why This Project Uses VSA + Machine Learning, explains why the strategy detector and learned filter have different roles and why training is completed in the backend. A walkthrough, selected learning videos and a quiz support active learning. The FAQ answers common beginner questions about guarantees, M5 data, confidence, WATCH_ONLY, news, MT5 and backtesting, while a downloadable glossary defines the main trading and ML terms. These pages make the system usable by people who cannot interpret raw metrics or code.

News-risk page and Forex Factory connection

The News Risk Filter reads the public weekly economic-calendar XML supplied through the Forex Factory/Faireconomy feed and focuses on relevant USD events such as CPI, non-farm payrolls and FOMC announcements. The page shows source status, broker-server UTC offset and upcoming events, with high-impact items highlighted. The gate is deliberately fail-safe: an event within the configured window blocks a recommendation, and an unavailable or unverified live calendar does not silently permit execution. A cached project calendar can keep the page informative, but it is labelled as a fallback and is not treated as verified execution evidence. News is therefore a safety control rather than a feature used to claim predictive accuracy.

The ML page loads cached artefacts and small result files. Full historical extraction, training, robustness evaluation and XAI-audit generation are offline tasks. This reduces startup work, provides stable recommendations between retraining runs and makes the training/advisor separation visible to an examiner. Cache boundaries are placed around immutable data reads rather than around live freshness checks.

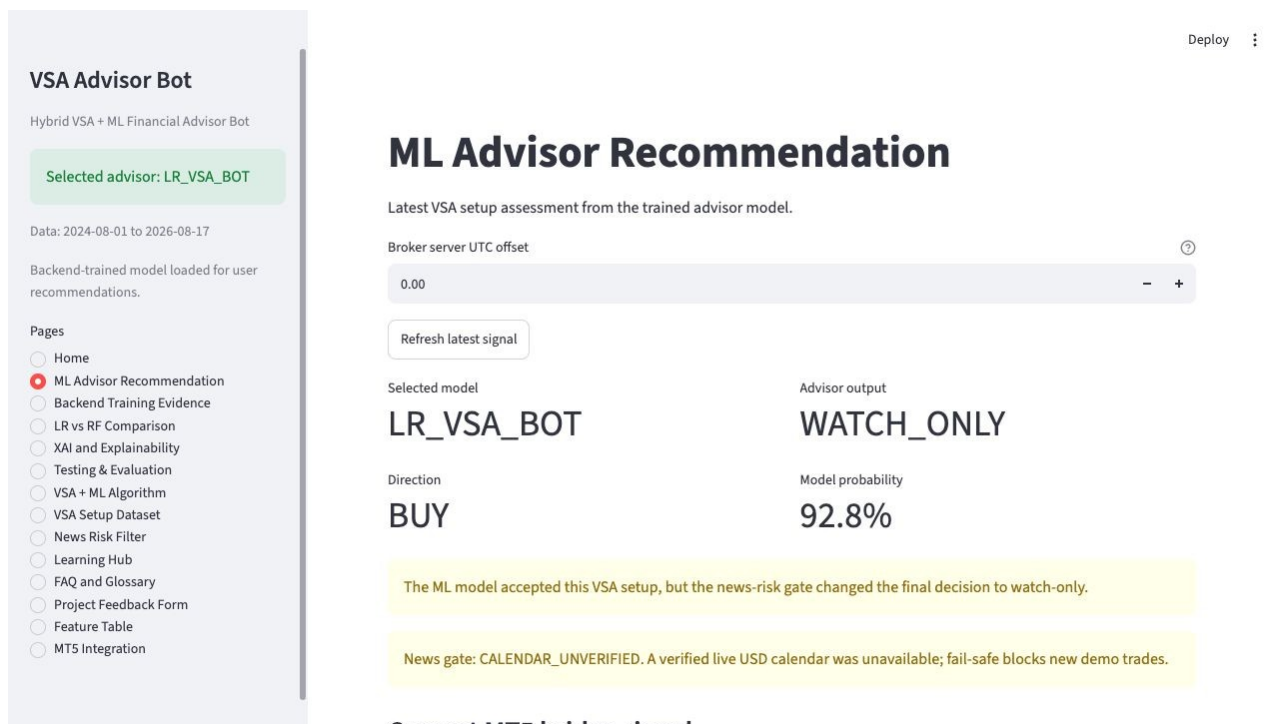


Figure 7. ML Advisor recommendation page reading the persisted model and latest compact signal artefact.
Source: author-captured prototype screen.

4.5 Faithful XAI implementation

For each unseen row, the implementation applies the fitted scaler, multiplies every standardised feature by its fitted LR coefficient, adds the intercept and applies the

logistic function. The audit stores the 24 contributions, reconstructed logit, reconstructed probability and model probability. The maximum discrepancy across all 65 unseen cases is 1.665×10^{-16} , below the 10^{-12} tolerance. This confirms computational faithfulness to the saved LR model, not economic causality or user comprehension.

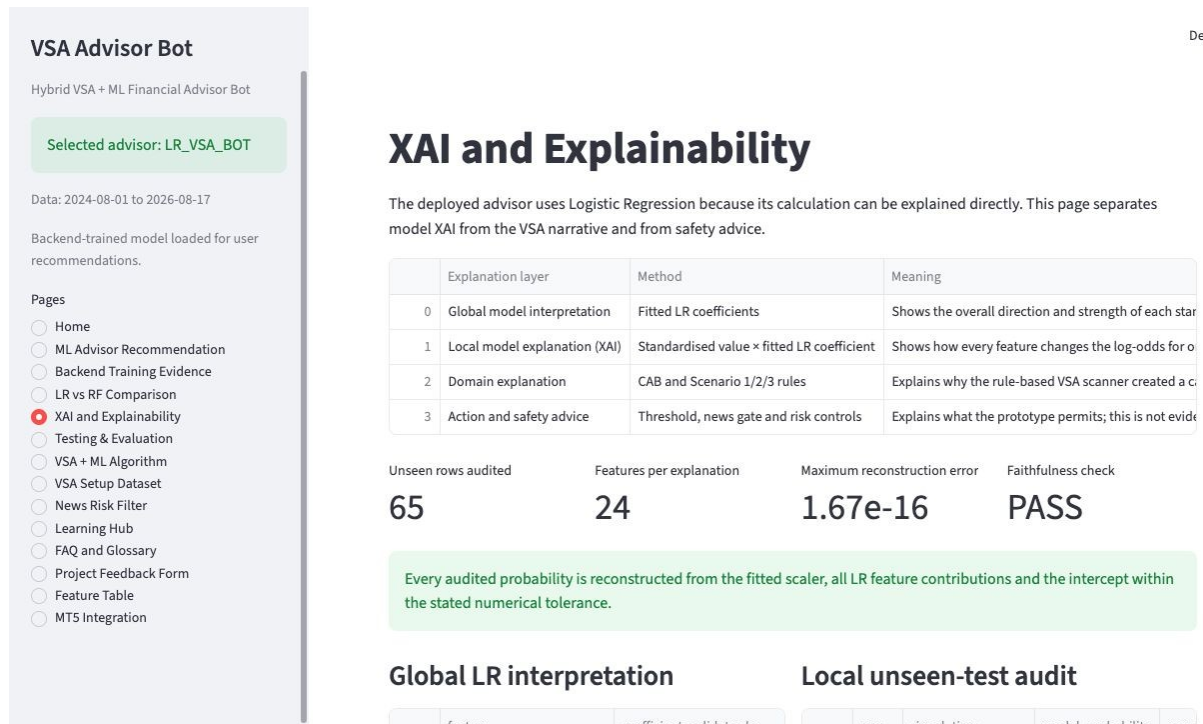


Figure 8. XAI page separating model attribution, domain explanation and safety advice; all unseen predictions pass the reconstruction audit. Source: author-captured prototype screen.

4.6 MT5 export, bridge and replay implementation

VSA_M5_Bar_Exporter_EA writes the latest 2,500 completed M5 bars to the Common Files area every ten seconds when a new closed bar is available. Python can then perform inference without relying on the forming candle. VSA_ML_Signal_Bridge_EA reads the current signal, validates its schema and rejects stale records. Desktop and push alerts are enabled separately from execution. VSA_LR_Backtest_Replay_EA replays the fixed out-of-sample signal file in Strategy Tester and can toggle UseMLFilter while holding the VSA setup, period and lot size constant.

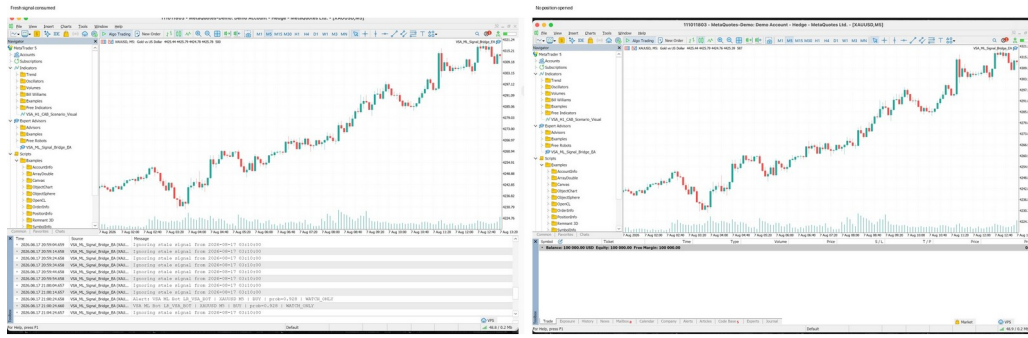


Figure 9. MT5 bridge evidence: fresh CSV consumed in WATCH_ONLY mode and empty Trade tab. Source: author-captured MT5 evidence.

4.7 MT5 visual scenario indicator

A separate MT5 visual indicator helps users inspect the rule detector directly on the chart. The indicator marks the CAB zone and its upper and lower trigger levels, then labels confirmed Scenario 1, 2 or 3 events with a directional arrow. It is read-only: it does not train a model, change the LR probability or open a trade. This separation is intentional. The indicator teaches the domain pattern and allows a user or examiner to compare the visible candle sequence with the Python/MQL5 specification, while the ML bridge remains responsible for the trained recommendation and safety state.



Figure 10. MT5 XAUUSD M5 visual indicator marking a buy-side CAB zone and confirmed Scenario 1 signal. Source: author-captured prototype screen.

Chapter 5 — Evaluation

5.1 Evaluation strategy

Evaluation is organised at four levels. Unit and integration tests examine software correctness. Chronological classification metrics assess the learned filter. R-based simulation and matched MT5 replays assess strategy behaviour. Robustness tests examine sensitivity to time, features and transaction costs. A target-user study will assess usability and explanation comprehension after consent; it is not yet reported. This layered design avoids treating one profitable backtest or one accuracy value as proof of a successful advisor.

Table 4. Evaluation framework mapped to the project aims.

Evaluation layer	Question	Evidence	Decision criterion
Software	Do components obey their contracts?	19 pytest tests; MT5 compile and bridge logs	All automated tests pass; invalid/stale signals fail closed
Predictive	Does ML discriminate valid VSA setups?	Development and unseen metrics, confusion matrices	Select on development F1; report test once
Trading	Does filtering change outcomes under fixed rules?	R simulation; matched MT5 replay	Compare VSA+LR with identical VSA-only baseline
Robustness	Are results stable to period, features and costs?	Walk-forward, bootstrap, cost sensitivity	Report dispersion and negative folds
Human factors	Can target users understand and use advice?	Consent-led questionnaire/tasks	Pending genuine participant data

Source: author's own work unless otherwise stated.

5.2 Software testing and integration evidence

The automated suite contains 19 passing tests. Coverage includes feature construction, chronological splitting, no-overlap assertions, model artefact compatibility, signal-schema fields, stale-signal handling, explanation reconstruction and mirrored strategy behaviour. The MT5 bridge and M5 exporter compile with zero errors and zero warnings. Runtime evidence shows an earlier missing-file state, a delimiter/header mismatch, stale-signal rejection and then successful consumption after correction. Keeping the failed states is useful engineering evidence because it shows the interface was tested at its boundaries, not only on a happy path.

The bridge test produced a desktop alert and mobile push notification for XAUUSD M5 with probability 0.928 in WATCH_ONLY mode. The contemporaneous Trade tab remained empty. This demonstrates that Python-to-MT5 communication worked while the default execution gate prevented a trade. It

does not demonstrate live profitability or order-management reliability; those require controlled demo-forward testing.

5.3 Predictive model results

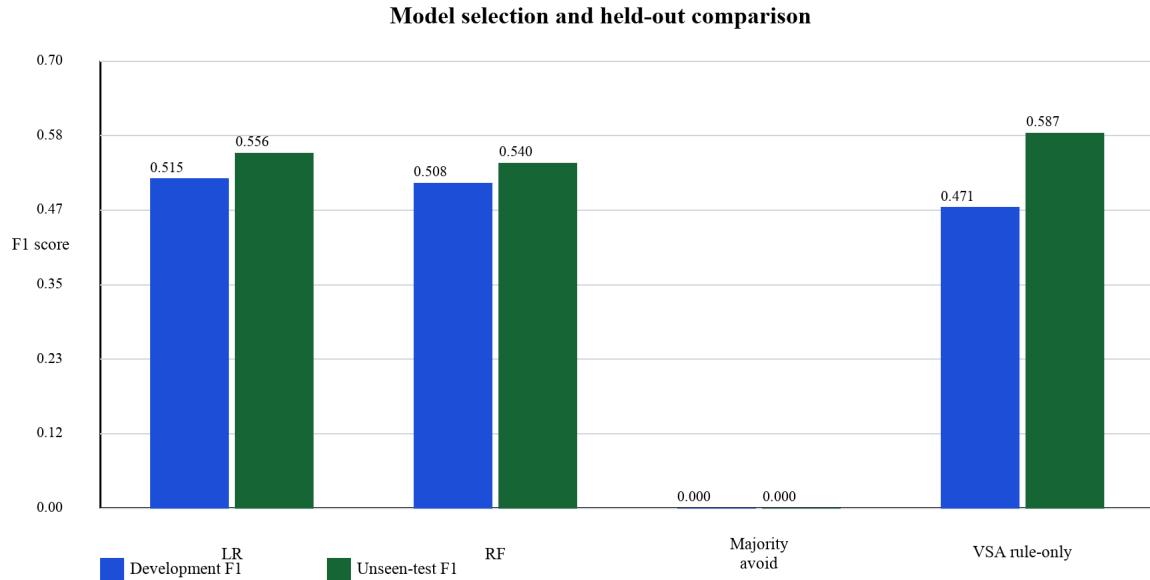


Figure 11. Development and unseen-test F1 for candidate models and baselines. Source: author’s own results.

Table 5. Classification performance on 65-row development and 65-row unseen-test sets.

Model	Threshold	Development F1	Test F1	Test precision	Test recall	Test balanced accuracy
Majority avoid	—	0.000	0.000	0.000	0.000	0.500
VSA rule-only	—	0.471	0.587	0.415	1.000	0.500
Logistic Regression	0.35	0.515	0.556	0.444	0.741	0.541
Random Forest	0.46	0.508	0.540	0.472	0.630	0.565
HistGradientBoo- sting (context)	development- tuned	0.524	0.542	—	—	—

Source: author’s own work unless otherwise stated.

LR is selected because it narrowly beats RF on the predefined development F1 criterion (0.515 versus 0.508) and offers direct local explanations. On unseen test data LR reaches F1 0.556 and RF 0.540. RF has slightly higher test precision and balanced accuracy, whereas LR accepts more true positives. The differences are small relative to the test size; the evidence supports LR as the project’s chosen transparent operating point, not a universal claim that LR is the better classifier.

The VSA rule-only baseline has a higher raw test F1 of 0.587 because it predicts every detected candidate as positive, giving recall 1.0 and balanced accuracy 0.5. This apparent contradiction is instructive: classification F1 is not a direct measure

of trading value. The learned filter rejects candidates, lowering recall but changing the distribution of realised outcomes. The report therefore evaluates the same predictions under the strategy payoff separately.

5.4 Strategy outcome and matched MT5 comparison

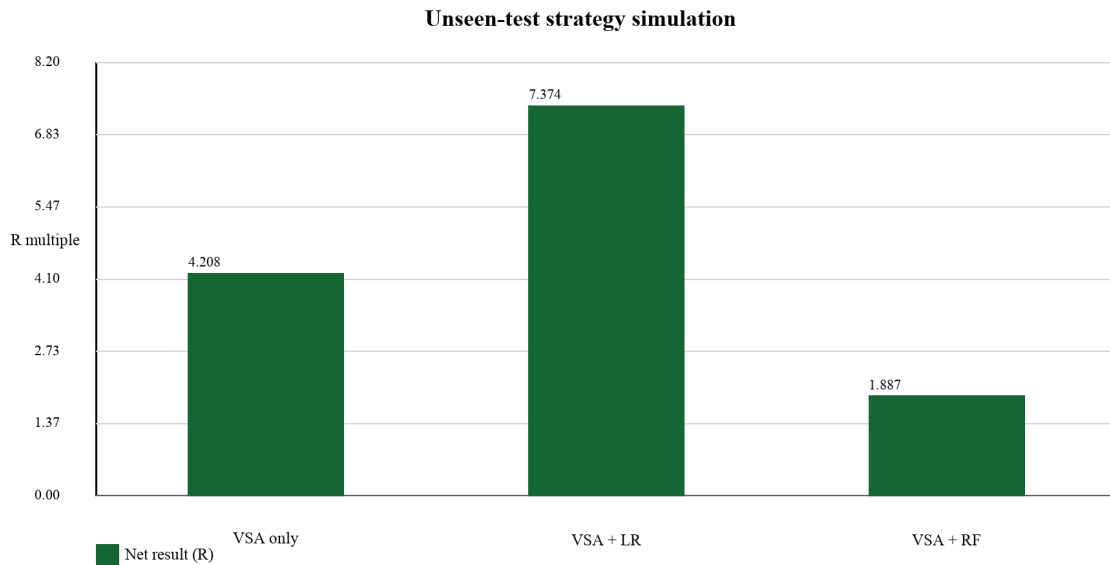


Figure 12. Unseen-test net result under the fixed VSA trade-management simulation. Source: author's own results.

Across the 65 unseen setups, VSA-only produces 4.208R. The LR filter accepts 45 setups (69.2%) and produces 7.374R; RF accepts 36 and produces 1.887R. The LR result is a 3.166R improvement over the unfiltered baseline under the implemented assumptions. Because setups may overlap and costs are estimated rather than broker-executed, R results are evidence of comparative behaviour, not a forecast of account return.

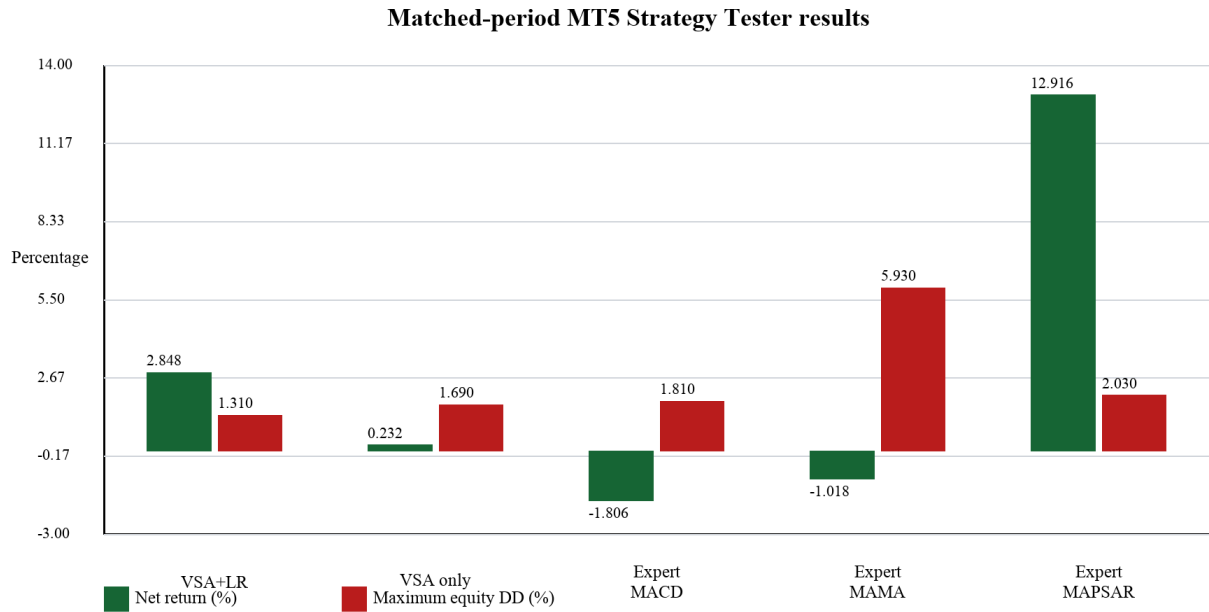


Figure 13. MT5 matched-period net return and maximum equity drawdown. Source: author's own Strategy Tester reports.

Table 6. Matched XAUUSD M5 Strategy Tester comparison, 1 June–17 August 2026, USD 10,000, 1:100, real ticks and 0.01 lots.

EA	Net profit	Return	Max equity DD	Profit factor	Sharpe	Trades	Win rate
VSA + LR	\$284.84	2.85%	1.31%	1.74	5.97	35	31.43%
VSA only	\$23.21	0.23%	1.69%	1.04	0.40	46	21.74%
ExpertMACD	-\$180.58	-1.81%	1.81%	0.30	-5.00	1,118	12.34%
ExpertMAMA	-\$101.80	-1.02%	5.93%	0.39	-0.39	22	90.91%
ExpertMAPSAR	\$1,291.65	12.92%	2.03%	15.13	6.02	80	96.25%

Source: author's own work unless otherwise stated.

The strongest internal comparison is VSA+LR against VSA-only, because the replay EA changes only the ML gate. LR increases net profit from \$23.21 to \$284.84, improves profit factor from 1.04 to 1.74 and reduces maximum equity drawdown from 1.69% to 1.31%, while taking fewer trades. This supports RQ1 for this test period.

The built-in EAs are contextual rather than causal baselines. ExpertMAPSAR performs best in this two-and-a-half-month window, while MACD and MAMA lose money. Their entries, exits and effective risk differ from the VSA system despite matched symbol, period, deposit, modelling and lot size. It would be misleading to claim that ExpertMAPSAR proves or disproves the learning design. The comparison instead shows that results are strategy- and regime-dependent and that one short period cannot establish superiority.

5.5 Robustness, limitations and threats to validity

Walk-forward evaluation is mixed: some chronological folds are positive and others negative. The mean result is therefore not stable enough for a profitability claim. A core ten-feature model and the full 24-feature model were also compared using bootstrap confidence intervals; the intervals for their difference cross zero. The study cannot conclude that all additional features improve generalisation. This supports the examiner's recommendation to validate a smaller core before expanding the strategy.

Transaction-cost sensitivity reduces outcomes as assumed spread/slippage increases. Exact simulated values depend on the replay convention, but the direction is consistent: marginal setups are the first to lose value. The MT5 comparison uses zero-latency Strategy Tester execution and should be interpreted as a controlled technical benchmark rather than a live execution estimate. Future evaluation should repeat the frozen pipeline across longer periods, multiple brokers, realistic variable spreads and a demo-forward window.

Internal validity is threatened by possible errors in translating discretionary VSA descriptions, overlapping setup horizons and threshold choices. The canonical table, mirrored tests, embargo and saved predictions mitigate but do not eliminate these issues. Construct validity is limited because `valid_trade` represents the implemented 1R/4R outcome, not general market correctness. External validity is limited to one instrument, timeframe and broker tick-volume feed. Statistical conclusion validity is limited by only 65 unseen setups. Any numerical advantage should therefore be reported with uncertainty and replicated.

5.6 Explainability evaluation

The LR faithfulness audit covers every unseen row and all 24 features. Maximum probability reconstruction error is 1.665×10^{-16} against a tolerance of 10^{-12} , producing PASS. This is stronger evidence than a selected example because it confirms the explanation formula over the complete test set. Global coefficients describe overall model direction; local contributions describe one prediction; the CAB narrative explains the rule detector; and action advice describes system policy. The interface labels these separately.

What remains unknown is whether a beginner understands the explanation, notices uncertainty or changes a risky decision. Faithfulness is a technical property, while usefulness is a human-factor property. Tomorrow's study should recruit 10–15 target users, obtain signed consent, assign a small set of recommendation-interpretation tasks and collect Likert and open-text responses. Planned outcomes

include task accuracy, confidence, explanation usefulness, risk comprehension and qualitative themes. Results must be reported even if they are negative.

Chapter 6 — Conclusion and Future Work

6.1 Conclusion

The project delivers a complete educational financial-advisor prototype that connects a formal VSA strategy, supervised learning, faithful model explanation, a non-technical web interface and MT5. It answers the template with a system that gathers financial data, trains outside the interface, persists evidence and generates advice that can be inspected. The three CAB scenarios and risk rules are explicit, mirrored and testable. LR and RF are trained and stored separately; LR is selected on development F1 and retained because its predictive result is competitive and its individual decisions can be reconstructed exactly.

On the unseen set, LR achieves F1 0.556 and the VSA+LR simulation yields 7.374R compared with 4.208R for unfiltered VSA. The matched MT5 replay shows the same directional conclusion: 2.85% versus 0.23% net return, with lower maximum equity drawdown. These findings support a learned VSA filter within the evaluated period. They do not establish future profitability. Mixed walk-forward folds, overlapping confidence intervals for feature sets and the strong period-specific ExpertMAPSAR result show why the conclusion must remain bounded.

The XAI contribution is equally important. The system does not use ‘explainable’, ‘advice’ and ‘XAI’ interchangeably. It separates model attribution, domain rationale and action policy, and verifies all unseen LR probabilities numerically. The integration also fails safely: a valid current CSV can trigger desktop and mobile advice while demo execution remains disabled and the Trade tab stays empty.

6.2 Future work

Before the 28 September submission, immediate work is to complete the consented target-user study, analyse genuine responses, update the evaluation chapter, record the required 3–5 minute MP4 and perform a final citation/evidence audit. These are submission requirements, not optional technical extensions.

After submission, the most valuable technical extension is a frozen demo-forward evaluation across several market regimes and broker feeds with measured spread, slippage and latency. Repeated walk-forward retraining should use a predetermined schedule and report uncertainty. Probability calibration and abstention could improve the meaning of confidence. A smaller feature set should be challenged against the full set on more data before new indicators are added. SHAP could

provide a common attribution method for nonlinear candidates, but it should be evaluated with users rather than added as a decorative chart.

A later execution prototype could enable demo orders with server-side risk caps, daily loss limits, duplicate-signal protection, partial-close verification and reconciliation after terminal restarts. Live-money trading should remain out of scope until those controls and a sustained demo-forward record exist. Multi-instrument testing, news-source redundancy and accessibility testing would improve external validity. The correct direction is deeper validation of a small, transparent system, not uncontrolled expansion.

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Appendix A — Report Word-Count Control

Table 7. Automated count of narrative text generated for the six chapters. Captions, tables, references and front matter are excluded, matching the stated exclusions where applicable.

Section	Calculated words
Chapter 1 — Introduction	867
Chapter 2 — Literature Review	1,179
Chapter 3 — Design	633
Chapter 4 — Implementation	1,046
Chapter 5 — Evaluation	943
Chapter 6 — Conclusion and Future Work	430
Total chapters	5,098

Source: author's own work unless otherwise stated.

Appendix B — Pending Target-User Study

Status at this draft: participant information, consent, questionnaire and analysis templates are prepared. Signed forms and responses will be inserted only after genuine collection. No participant result is claimed in this version.

Appendix C — Demonstration and Evidence Checklist

- Backend training terminal and generated artefacts.
- VSA setup dataset and 24-feature table.
- Separate LR/RF development and unseen-test comparison.
- Selected LR model and faithful local explanation.
- Streamlit recommendation, news-risk gate and learning pages.
- MT5 M5 exporter, fresh CSV consumption, desktop/mobile alert and empty Trade tab.
- Strategy Tester VSA+LR versus VSA-only plus contextual built-in EAs.
- User-study evidence after signed consent, followed by final limitations.

Required video: 3–5 minutes, MP4 format, showing the end-to-end prototype and evidence rather than only slides.